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Where the **agentic stack** is being built.

Lost in Conversation: Why Voice Agents Break When Users Interrupt

Irvin Cardoza | AGNTCon + MCPCon China | 2026

What does interruption actually mean?



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user

Reschedule my 2 PM interview with Maya to 4 PM, book a room, and email her the update.

agent

Sure — moving the meeting to 4 PM, booking a room, and notifying Maya...

user — interrupts

Wait — make it virtual instead, and don't email her yet.

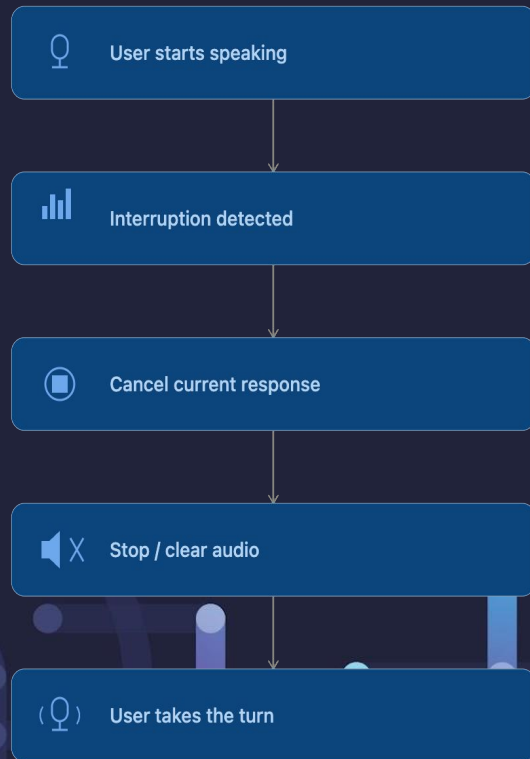
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Barge-in is mostly solved

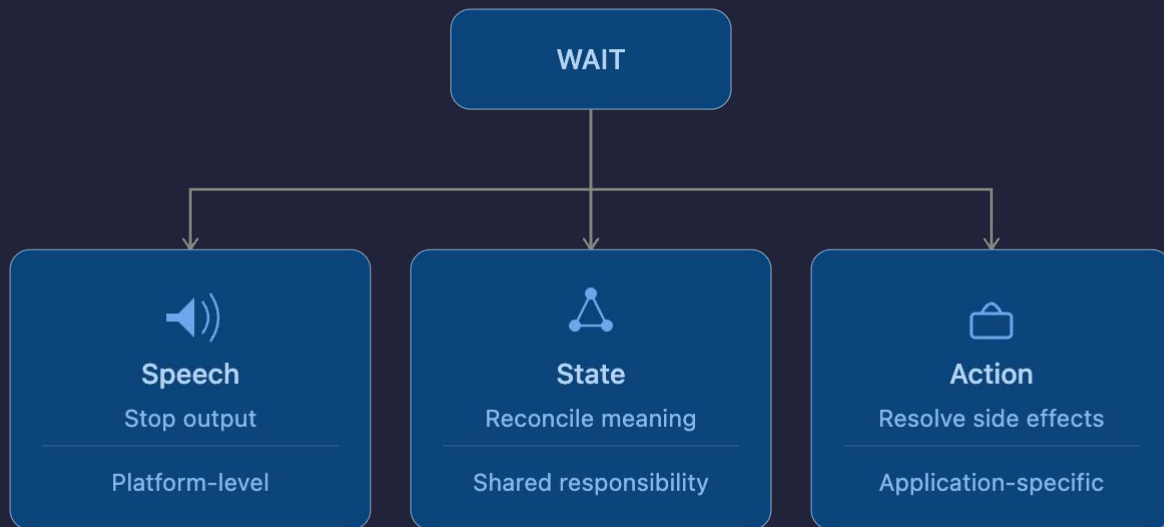
Modern voice stacks already handle this well.

For example:

OpenAI Realtime · Gemini Live · LiveKit



Stopping speech is easy. Knowing what should stop is hard.



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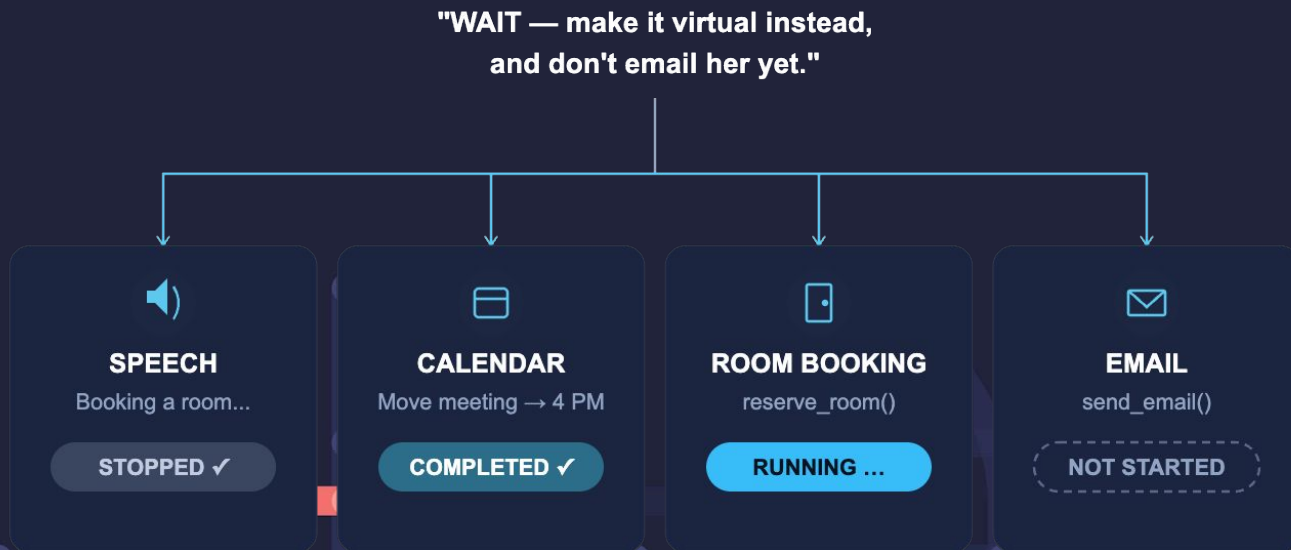


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Concurrency

When a user interrupts, speech, state, and tool execution may all be at different points.



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Three Truths

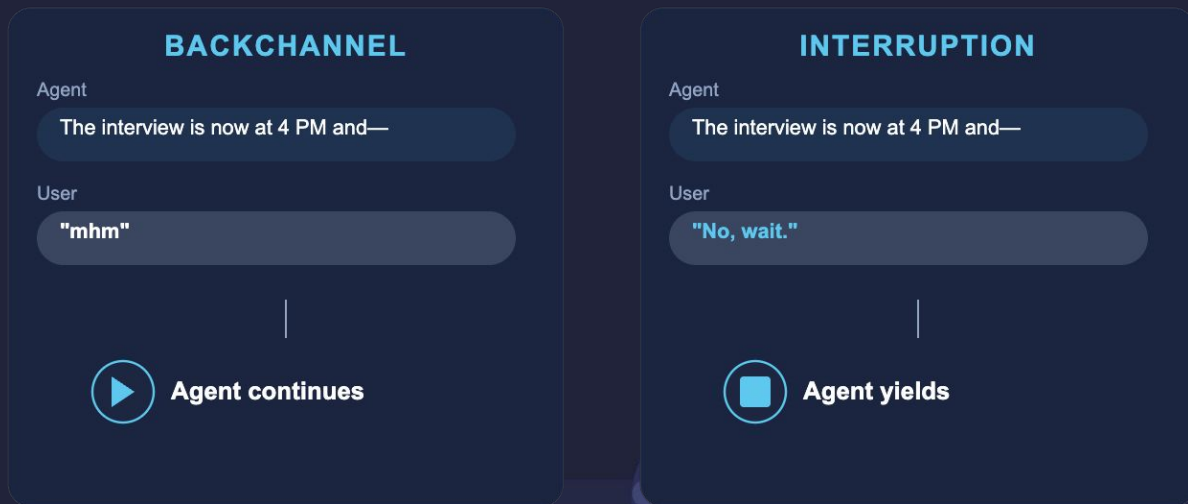
After an interruption, the agent must reconcile three different realities.

INTERRUPTION



Truth 1- Intent

Not every time the user speaks over the agent means they want to interrupt.



Speech detected $\neq$ turn requested

Truth 2 - What Did They Hear?



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The model may generate more than the user actually hears before interrupting.

MODEL GENERATED

I moved the interview to 4 PM, booked a room, and emailed Maya.

USER ACTUALLY HEARD

I moved the interview to 4 PM and—



Generated

Synthesized

Played

Generated output ≠ delivered conversation

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Reconciliation

- Stop the response immediately when the user interrupts.
- Track what the user actually heard, not everything the model generated.
- Update conversation state so the next turn starts from the correct shared context.

Truth 3: What Has Happened?

- A tool may already be running when the user interrupts.
- Stopping the conversation does not guarantee the external action stops.
- If the action has already completed, recovery may require a separate compensating action.

Tool State

The right interruption response depends on where the action is in its lifecycle.

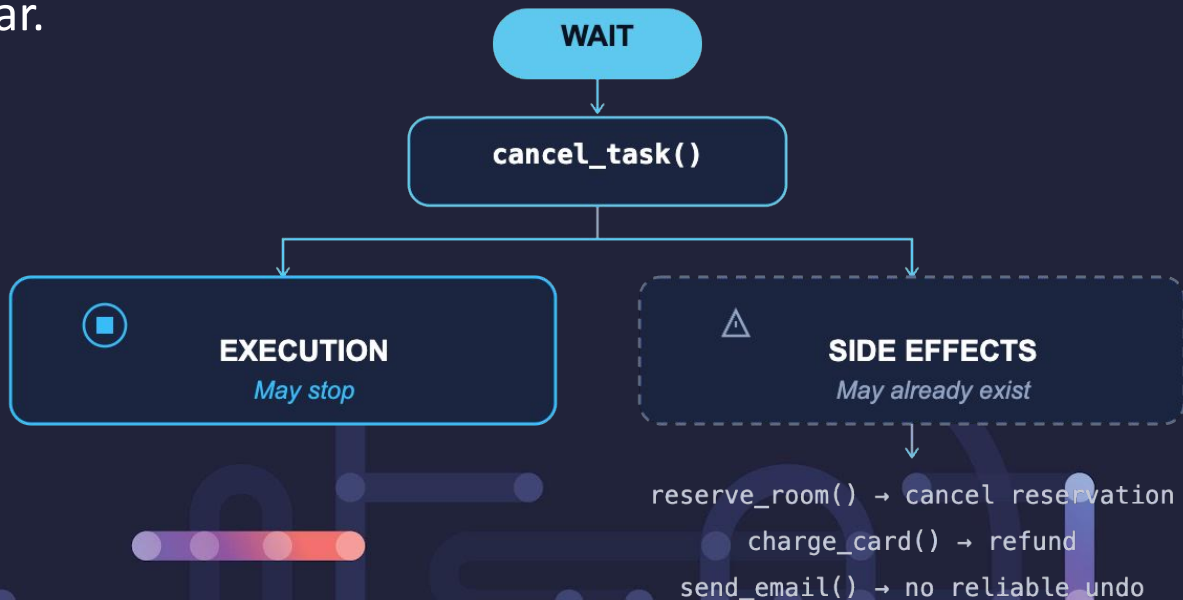


After commitment, interruption becomes recovery.

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Cancellation ≠ Rollback

Stopping a task does not guarantee that its real-world effects disappear.



Cancellation stops execution. Recovery repairs state.

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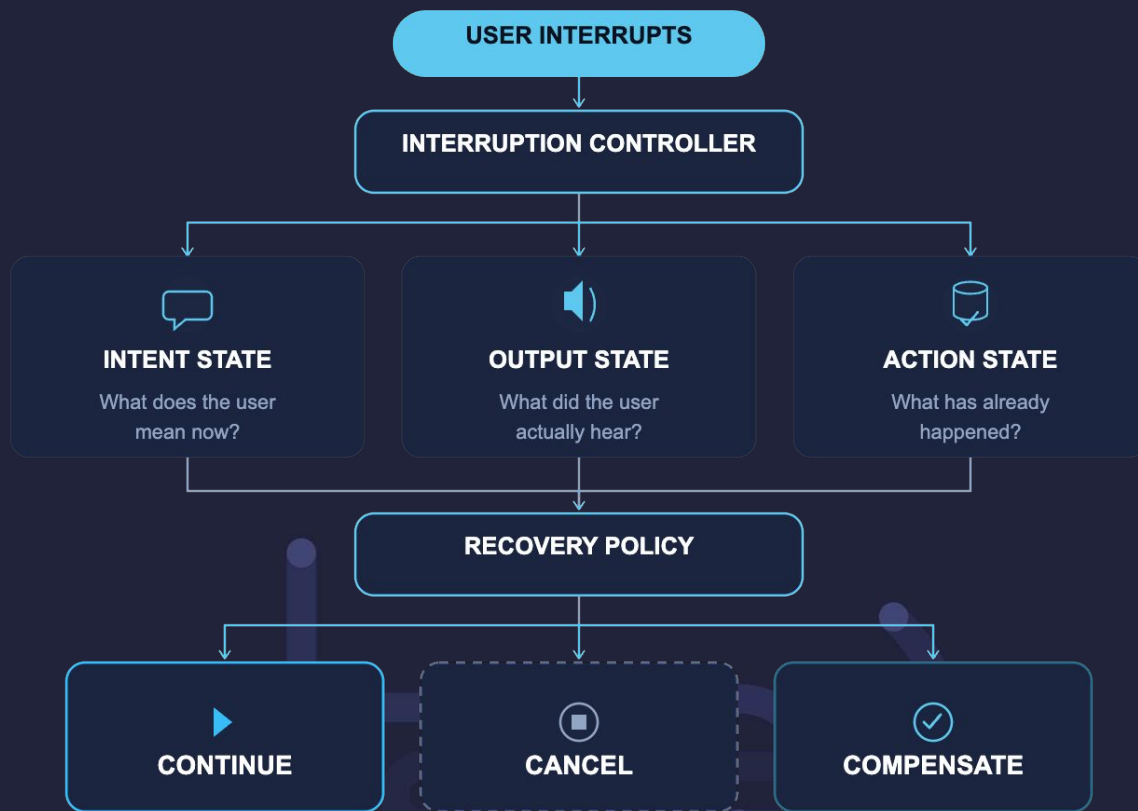
Interruption Contract

- A design rule for what the agent must resolve whenever a user interrupts.
- It requires the agent to reconcile intent, delivered output, and action state before continuing.
- It gives the system a consistent basis for whether to continue, cancel, or compensate.



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Takeaway

Stopping speech is a UX behavior. Recovering from interruption is a systems property.

Reliable agents don't just stop - they recover coherently.

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An abstract graphic at the bottom of the slide featuring a network of horizontal and vertical lines in shades of blue, purple, and orange. Small circles of the same colors are placed at various points along these lines, creating a sense of a complex, interconnected system or data flow.



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The agent must know what happened before it can recover.

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Click to edit title

- Click to edit text
 - Second level
 - Third level
 - Fourth level
 - » Fifth level



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